

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (CCE) (Credit System) Degree Examinations**

May - June 2023

20CC603 - INTERNET OF THINGS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) With neat diagram explain general architecture of IoT system and discuss various features of it.	10	L*2	1	1,2
	b) What are the design principles and needed capabilities of IoT architecture? Discuss.	10	L2	1	1,2
2.	a) List the devices that are identified with IoT system. What are the major tasks of device management in IoT system?	10	L2	1	1,2
	b) How Sensors are different from Actuators? Classify and characterize Sensors and Actuators that are used in IoT system.	10	L2	1	1,2
3.	a) What is cloud computing? Discuss its role in IoT system.	10	L2	2	1,2
	b) Discuss various components and peripherals of Raspberry Pi.	10	L2	2	1,2
Unit – II					
4.	a) Compare and contrast High-Level ZigBee and ZigBee IP Protocol Stack.	10	L2	3	1,2
	b) What Is MQTT? Discuss working of MQTT protocol.	10	L2	3	1,2
5.	a) Discuss various types of messages supported by CoAP. List features of CoAP protocol.	10	L2	3	1,2
	b) Discuss the challenges and solution measures of IoT data analytics.	10	L2	4	1,2
6.	a) Explain Data generation, Data acquisition and Data validation in IoT system.	10	L2	4	1,2
	b) Discuss cloud computing paradigm and its services in IoT systems.	10	L2	4	1,2
Unit – III					
7.	a) Discuss any 5 IoT applications in brief.	10	L2	5	1,2
	b) Outline IoT home automation system and discuss components associated with the system.	10	L3	5	1,2
8.	a) Discuss IoT application requirements and capabilities for an industrial application.	10	L2	5	1,2
	b) Outline any Use-Case of IoT in smart cities.	10	L3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
